

NUCLEAR ENERGY TECHNOLOGIES DESIGN COMPETITION PRELIMINARY EVALUATION REPORT

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Category Name: Detailed Design

Team Type: School Team

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**PRELIMINARY EVALUATION REPORT (PER)****TEAM CATEGORY**

BASE CATEGORY	
Detailed Design – 40 MWe Modular PWR	✓
Conceptual Design – Modular Reactor (SMR/MMR)	
Conceptual Design – Research Reactor (RR)	
SUB-CATEGORIES SYSTEM DEVELOPMENT	
Reactor Core Design	
Nuclear Fuel and Waste Management	
Energy Cycle Design	
Passive Safety System Design	
Nuclear Instrumentation and Control System Design	
APPLICATION CATEGORY	
Software Development	
Robotic System Design	

TEAM TYPE

School Team	✓
Professional Team	
Undergraduate Team	



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ABBREVIATIONS

AHEC – Autonomous Hydrogen Energy Complex
AHP – Analytical Hierarchy Process
BOL – Beginning of Life
CFD – Computational Fluid Dynamics
CVCS – Chemical and Volume Control System
DCS – Distributed Control System
DiD – Defence in Depth
EOL – End of Life
EPZ – Emergency Planning Zone
FOM – Figure of Merit
GP – Gaussian Process
HPC – High-Performance Computing
I&C – Instrumentation and Control
IAEA – International Atomic Energy Agency
iPWR – Integral Pressurized Water Reactor
LHTES – Latent Heat Thermal Energy Storage
MDNBR – Minimum Departure from Nucleate Boiling Ratio
MOC – Middle of Cycle



NPP – Nuclear Power Plant
PCT – Peak Cladding Temperature
PWR – Pressurized Water Reactor
RANS – Reynolds-Averaged Navier–Stokes
REE – Rare Earth Element
RPV – Reactor Pressure Vessel
SBF – Soluble-Boron-Free
SMR – Small Modular Reactor
SOE – Solid Oxide Electrolyzer
TCES – Thermochemical Energy Storage
TES – Thermochemical Energy Storage
TH – Thermal-Hydraulics
wFOM – Weighted Figure of Merit
3S – Safety, Security, and Safeguards



1. PROJECT ABSTRACT / SCOPE OF THE PROJECT

1.1. Project Abstract

The Aegis-40 is a 40 MWe integral Pressurized Water Reactor (iPWR) designed for modular, secure, and economically competitive deployment. The design combines a soluble-boron-free (SBF) core, open-source neutronics and thermal-hydraulic modeling (OpenMC, OpenFOAM), and an integrated multi-application energy system.

Key system-level innovations include: (i) Thermochemical Energy Storage (TES) for district heating, (ii) Solid Oxide Electrolysis (SOE) for off-peak hydrogen production, (iii) a Digital Twin-based Instrumentation and Control (I&C) architecture for predictive monitoring, and (iv) a Weighted Figure of Merit (wFOM) framework to guide design optimization.

The project follows a fully integrated Safety, Security, and Safeguards (3S) approach and targets high passive safety performance, operational flexibility, and strong alignment with Turkey's energy and industrial strategy.

1.2. Scope of the Project

This project falls within the Detail Design category of the TEKNOFEST 2026 Nuclear Energy Technologies Design Competition. The scope encompasses: (i) detailed neutronics core design and fuel management; (ii) thermal-hydraulic analysis of steady-state and transient behaviour; (iii) Instrumentation and Control (I&C) system architecture with a Digital Twin surrogate model; (iv) nuclear power plant (NPP) layout and site integration for a Black Sea coastal deployment; (v) integration of non-electrical applications — district heating via TES and hydrogen production via SOE during off-peak hours; and (vi) comprehensive 3S analysis. All technical work is performed in accordance with IAEA safety standards (SSR-2/1 Rev. 1) and international nuclear engineering best practices.

2. TEAM INTRODUCTION

The project team is structured to efficiently address the multidisciplinary requirements of a 40 MWe Pressurized Water Reactor (PWR) detailed design. Task distribution is presented schematically in Fig. 1, where responsibilities are grouped into interconnected domains to ensure both depth of expertise and effective integration. A centralized management structure oversees coordination between subsystems, while dedicated leads in each domain guide technical development and ensure consistency with project objectives.

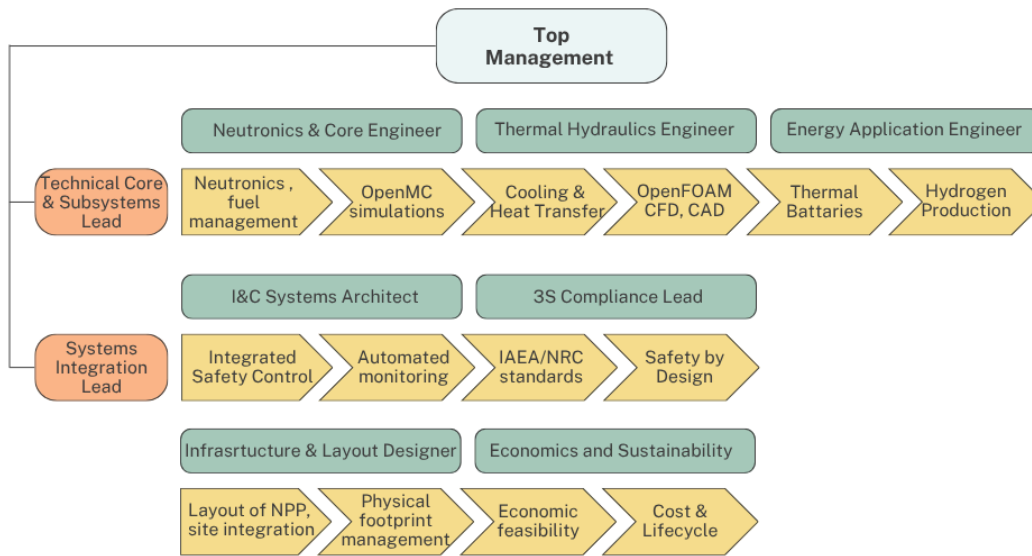


Fig. 1. Schematic representation of the project team structure and task distribution for the 40 MWe PWR detailed design.

The diagram in Fig.1 illustrates the hierarchical organization under top management, divided into two primary technical domains: (i) core design and thermal-hydraulics, (ii) safety, instrumentation and control (I&C), regulatory compliance (3S), and infrastructure, layout, and sustainability. Each domain is associated with specific engineering responsibilities, including neutronics analysis using OpenMC, thermal-hydraulic modeling with OpenFOAM, safety-by-design implementation, and economic feasibility assessment. The workflow arrows indicate the sequential and integrated nature of tasks within each subsystem, while designated leads ensure coordination between technical areas.

The core team consists of:

- *Project Lead & Neutronics Engineer*
- *Thermal-Hydraulics Engineer*
- *I&C Systems Engineer and 3S Compliance Lead*
- *Energy Application Engineer*
- *Infrastructure & Layout Engineer*
- *Economics & Sustainability Analyst*

Each member is responsible for a defined subsystem, as illustrated in Fig. 1.

3. LITERATURE REVIEW

3.1. Introduction and Comparative Survey of iPWR Designs

Small Modular Reactors (SMRs) offer scalable, low-carbon power generation with reduced upfront capital costs compared to conventional large plants, leveraging factory-based modular construction to shorten build timelines and manage financial



risk [1]. The integral Pressurized Water Reactor (iPWR) architecture — where the reactor core, steam generators, and pressurizer are all housed within a single pressure vessel — has emerged as the most commercially mature SMR class [2]. The Korean SMART reactor (100 MWth) earned the world's first standard design approval for an integral PWR in 2012, demonstrating that internalising all primary components eliminates large-break loss-of-coolant accidents [2]. The Argentine CAREM-25 (32 MWe) — the primary reference for this project — demonstrates natural-circulation primary cooling without active reactor coolant pumps, substantially enhancing passive safety at low power densities [1, 5]. NuScale's Power Module (77 MWe) pioneered passive cooling via a shared underground pool and reduced the emergency planning zone to the plant boundary [2, 3]. The Korean i-SMR advances the state of the art with a boron-free core and in-vessel control rod drive mechanism, targeting core damage frequencies 1000 times lower than Generation III plants [2]. Reducing reactor power capacity increases the surface-area-to-volume ratio, enhancing passive decay heat removal by more than 233% relative to a 1000 MWe reference [2]. These precedents affirm the integral configuration as the appropriate architectural choice for the Aegis-40.

3.2. Core Neutronics and Reactivity Control

Criticality analysis, neutron flux distribution, reactivity coefficient characterisation, control rod worth, and burnup evolution must all be rigorously quantified to ensure safe, predictable core behaviour throughout the fuel cycle [4]. Neutron spectrum and enrichment must be co-optimised with fuel assembly geometry to maintain negative temperature and void reactivity coefficients in all operating states [5]. Modern iPWR designs increasingly pursue soluble-boron-free (SBF) operation to eliminate the Chemical and Volume Control System (CVCS) burden and reduce radioactive waste generation. Kim et al. [6] confirm that gadolinia (Gd_2O_3) burnable absorbers are practical candidates for full SBF operation in 40 MWe-class cores, providing excess reactivity control early in the cycle while Erbium (Er) maintains a flatter reactivity profile at end-of-life.

3.3. The 3S Pillars: Safety, Security, and Safeguards

The Aegis-40 implements Defence-in-Depth (DiD) through explicit architectural decisions. The reactor is placed below grade, which simultaneously enhances passive heat removal through natural convection pathways, provides inherent protection against external hazards (e.g., aircraft impact), and reduces the exposed security perimeter. Physical barriers correspond directly to DiD layers: fuel matrix and cladding, integral primary vessel, and compact containment [7, 8].

Security-by-Design is achieved through a reduced site footprint and a hardened underground layout, limiting unauthorized access points. Cybersecurity is implemented via a segregated I&C architecture, where the safety system is isolated by an air gap and protected by a one-way data diode, forming an independent defensive layer [9, 10].



Safeguards-by-Design is supported through the SBF core, which eliminates soluble boron handling systems and reduces operational complexity in nuclear material accounting [7, 9]. The 18-month fuel cycle minimizes refuelling frequency, aligning with international inspection and non-proliferation objectives.

Thus, each 3S pillar [12] is directly embedded into the physical and operational design of the Aegis-40 rather than treated as an external requirement.

3.4. Computational Methods: OpenMC and OpenFOAM

OpenMC is an open-source Monte Carlo particle transport code supporting continuous-energy cross sections, full 3D core modelling, eigenvalue criticality, and burnup/depletion tracking via its Python-C++ interface [13]. The Chebyshev Rational Approximation Method (CRAM) solves isotopic transmutation without numerical instability [14]. Its compatibility with ENDF/B-VIII.0, JEFF-3.3, and JENDL-4.0 cross-section libraries, combined with native stochastic uncertainty quantification, makes it uniquely suited to SMR sensitivity analyses [4, 13]. Ez Aldeen et al. [3] validated OpenMC against a NuScale-like SMR benchmark, directly confirming its applicability to the 40 MWe design class.

OpenFOAM is an open-source multi-physics CFD framework supporting custom nuclear solvers, buoyancy-driven natural convection (buoyantSimpleFoam), conjugate heat transfer, and two-phase flows via the foamForNuclear platform [18, 20, 25]. The κ - ω SST turbulence model is employed throughout [21, 23], and Zhang and Sanchez-Espinoza [19] validated a hybrid 3D-RPV/1D-secondary-loop approach for SMART-like reactors that forms the direct basis of the Aegis-40 TH methodology.

3.5. TES, District Heating, and Night-Time Hydrogen Production

Three primary categories of energy storage have been explored for nuclear power plant (NPP) integration: sensible heat storage, latent heat thermal energy storage (LHTES) using phase change materials (PCM), and thermochemical energy storage (TCES). Each presents trade-offs in energy density, temperature range, scalability, and economic feasibility.

LHTES systems exploit phase-change enthalpy but are often constrained by scale. Yurin et al. [29] analyzed a PCM-based system using a NaNO_3 – NaOH mixture, requiring over 11,000 tonnes of material and extensive heat exchanger infrastructure, resulting in a high specific capital cost ($\sim \$3,781/\text{kW}$). While thermally stable, such large-scale systems are poorly suited to modular reactor deployment.

In contrast, hydrogen-based energy storage systems provide greater flexibility. The same study evaluated an autonomous hydrogen energy complex (AHEC), demonstrating a significantly lower capital cost and improved economic performance. Complementary work by Milewski et al. [28] shows that Solid Oxide Electrolyzers (SOE) can be directly integrated into PWR steam cycles—particularly at the deaerator inlet—allowing hydrogen production with minimal system modification and increased operational flexibility.



Thermochemical energy storage based on reversible sorption reactions (e.g., zeolite materials) offers an alternative pathway, combining relatively high energy density with negligible long-term heat loss. These characteristics make TCES particularly attractive for district heating and seasonal storage applications in SMR systems.

Literature demonstrates that sorption-based TES and SOE hydrogen production can enhance SMR flexibility and economic performance. These concepts form the basis for the Aegis-40 integration described in Section 4.

3.6. Digital Twin Technology for Nuclear Applications

A Digital Twin integrates real-time sensor data with physics-based surrogate models to monitor plant safety margins and predict future behaviour. Hergesheimer et al. [17] demonstrate that physics-informed Gaussian process surrogate models trained on high-fidelity simulation data can replicate full-system responses in milliseconds, enabling real-time control-room integration. For the Aegis-40, this is particularly important for remote deployment scenarios where on-site expert staffing is limited. The surrogate receives live sensor inputs from the distributed control system and continuously estimates core power distribution, PCT, and safety margins, triggering automated alerts when any parameter approaches its design limit [17].

3.7. Turkey's Nuclear Energy Context and the Sinop Site

Turkey imports approximately 75% of its total energy, with roughly one-third of its natural gas from a single external supplier [27]. In 2024, coal (~35%) and natural gas (~19%) dominated electricity generation, with nuclear contributing nothing [27]. Four VVER-1200 units are under construction at Akkuyu and the Sinop site on the Black Sea is designated for the country's second plant. Kurt [26] formally identified Inceburun–Sinop as the optimal NPP location through multi-criteria analysis across geological, social, and cooling water criteria. The Sinop environmental impact assessment was approved by Turkey's Ministry of Environment and Urbanization in September 2020, and EUAS established a nuclear-focused subsidiary (TUNAS) in 2023 to develop the project [27].

4. METHODOLOGY AND FEASIBILITY

4.1. Methodology

The Aegis-40 design follows a staged, iterative systems-engineering workflow comprising five sequential phases. Phase 1 establishes the neutronics baseline using OpenMC. Phase 2 performs thermal-hydraulic analysis using OpenFOAM. Phase 3 couples the two codes iteratively to capture physics feedback. Phase 4 integrates and sizes the non-electrical applications — TES for district heating and SOE for night-time hydrogen production. Phase 5 constructs the Digital Twin surrogate model and integrates it with the I&C architecture. All phases are governed by the 3S framework from the outset. See Figure 2 for the methodology flowchart.

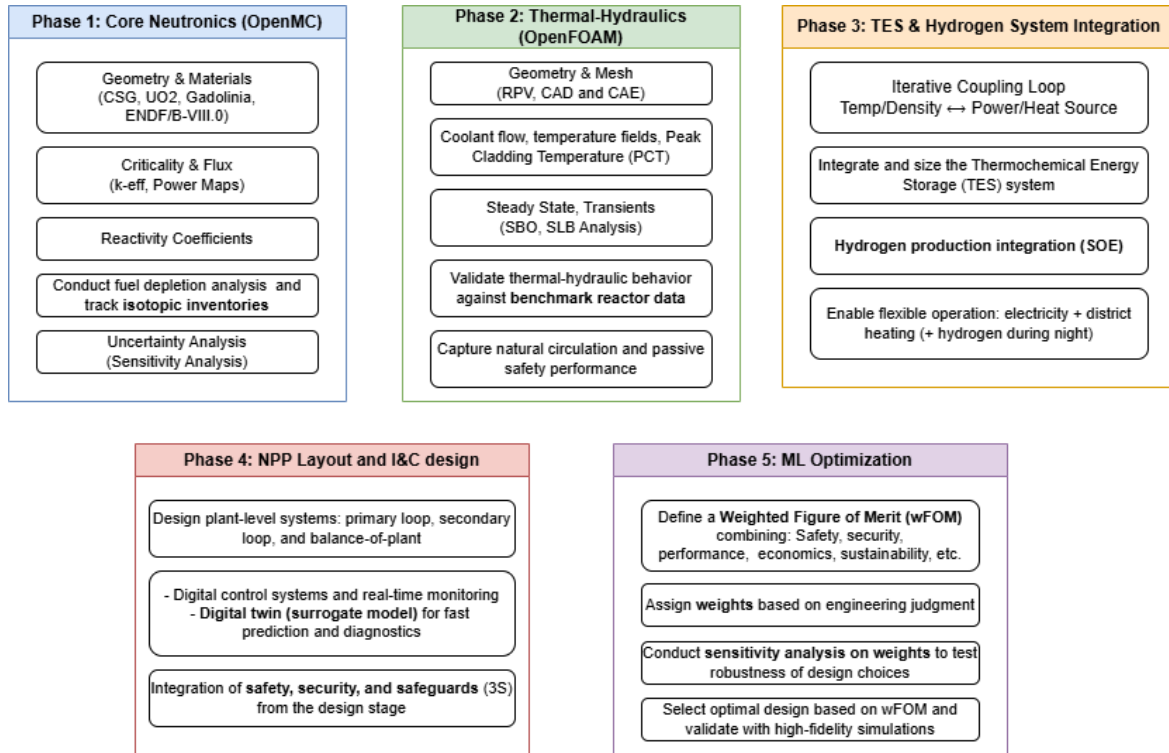


Fig. 2. Aegis-40 design methodology workflow from core analysis to system integration and optimization.

4.1.1 OpenMC

Neutronic modeling of the reactor core is performed using the Monte Carlo method with continuous-energy cross sections in the OpenMC v0.15+ code [13]. The geometric model is constructed parametrically via the Python API and includes fuel assemblies arranged in a standard lattice, with the capability for radial enrichment zoning and the incorporation of burnable absorbers.

The specific enrichment values, zoning configuration, and burnable absorber compositions will be determined through an iterative optimization process aimed at achieving target parameters such as criticality, power distribution uniformity, and fuel cycle length. Criticality is evaluated through the calculation of the effective multiplication factor k_{eff} , using an adaptive number of particles per batch and batches to ensure a statistical uncertainty below 0.001 [3].

The evolution of the isotopic composition is modeled using a depletion solver with the CRAM48 algorithm, which integrates the Bateman equations for more than 400 nuclides based on the ENDF/B-VIII.0 nuclear data library [14]. Reactivity temperature coefficients (Doppler and moderator) are calculated by performing a series of simulations at different temperatures, with interpolation accounting for Doppler broadening of resonance cross sections.

A preliminary two-zone configuration is assumed for workflow verification purposes (inner: $\sim 3.4\%$, outer: $\sim 4.0\%$), subject to revision following full optimization.

As part of the design preparation phase, a preliminary OpenMC model has been developed to establish a baseline geometry and verify simulation workflow functionality. The parameters used in this initial model — including assembly count, core radius, and shielding dimensions — are indicative and will be refined iteratively as the detailed design progresses. The results presented here are intended to confirm methodological validity and provide directional insight, not to represent final design specifications.

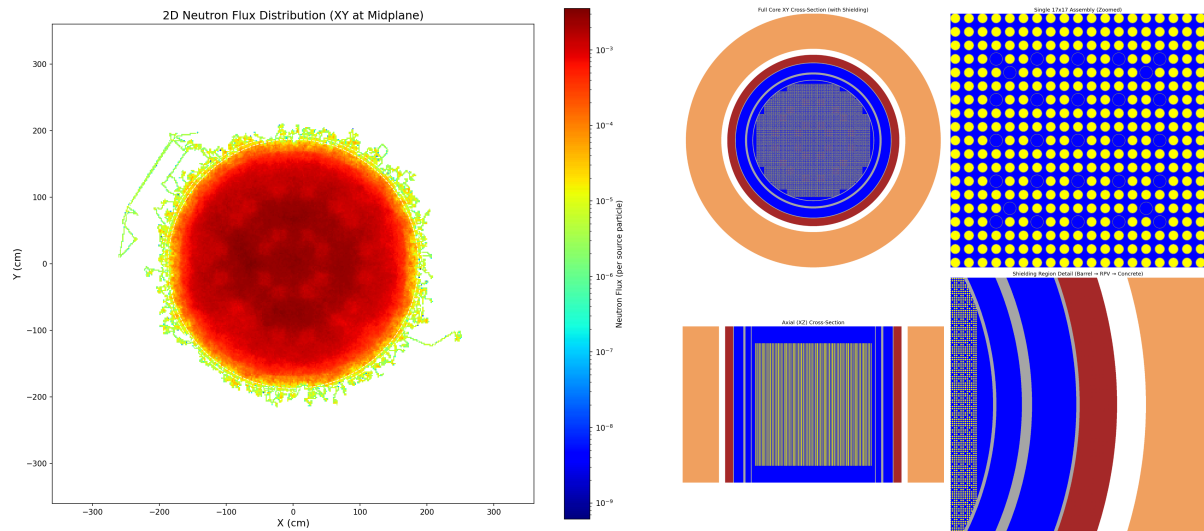


Fig. 3. OpenMC 2D neutron flux distribution and geometric model of the Aegis-40 core (preliminary)

4.1.2 OpenFOAM

This project employs a high-fidelity CFD approach implemented in the OpenFOAM finite volume framework. The workflow begins with pre-processing, utilizing detailed CAD geometries and the snappyHexMesh utility to generate unstructured meshes with specific boundary layer refinement to explicitly resolve fuel rods, cladding, and coolant channels. Solver execution leverages MPI parallelization for high-performance computing scalability, while post-processing is conducted via ParaView [18, 20].

The simulation solves the Favré-averaged Navier–Stokes equations for the conservation of mass, momentum, and energy [23]. Conjugate Heat Transfer is resolved at solid-fluid interfaces to accurately model the thermal interaction between the fuel pins and the coolant [20, 23]. Turbulence is addressed using RANS with the k-omega SST model, ensuring high-fidelity blending between bulk flow and boundary layers. The PIMPLE algorithm and second-order discretization schemes are employed to ensure numerical stability and accuracy [18].



Where boiling occurs, a six-equation Euler–Euler two-phase model is implemented [20]. Dimensionless numbers (Re, Pr, Pe) guide modeling assumptions and the evaluation of safety-critical parameters, including Critical Heat Flux and Departure from Nucleate Boiling Ratio [19, 20]. Due to the computational intractability of full-core CFD, a representative single fuel assembly is modeled to balance high-resolution accuracy with available resources [18, 22, 24].

The OpenFOAM results are iteratively coupled with OpenMC neutronic calculations to capture feedback effects between temperature, density, and reactivity.

4.1.3 Non-Electrical Applications

During daytime peak demand hours, secondary steam at $\sim 280^{\circ}\text{C}$ / 15 bar is diverted to a zeolite 13X TES module. Desorption (charging) occurs at $150\text{--}180^{\circ}\text{C}$; adsorption (discharging) releases heat at $80\text{--}90^{\circ}\text{C}$, matched to Turkish district heating supply temperatures. The module is sized for a ~ 350 GJ daily storage capacity (~ 390 tonnes of zeolite) to bridge peak winter demand of approximately 15 MWth, allowing the reactor to operate at constant 125 MWth without load-following [15, 16, 29].

During off-peak night-time hours ($\sim 22:00\text{--}06:00$), an $\text{O}^2\text{-SOE}$ unit is coupled to the deaerator steam inlet — the minimally invasive integration validated by Milewski et al. [28]. Operating at $\sim 740^{\circ}\text{C}$ with 5 MWe input, the SOE produces approximately 330 kg H_2 per 8-hour shift (~ 120 t/year) at 37.55 kWh/kg H_2 . Hydrogen is separated from the steam- H_2 mixture at the cathode outlet via a condenser-chiller train, yielding compressed H_2 at ~ 30 bar for industrial supply or on-site storage. Based on the Yurin et al. [29] benchmark, the electrolyzer-compression subsystem has an estimated specific capital cost of $\sim \$1,981/\text{kW}$, giving a total subsystem investment of approximately $\$10\text{--}15$ million with payback within 25 years [29].

4.1.4 Digital Twin and I&C Architecture

The Digital Twin is built in two stages. First, a simulation dataset of 200–400 OpenMC and OpenFOAM snapshots covering full power, partial power, and post-trip scenarios at BOL, MOC, and EOL is generated. A Gaussian process (GP) surrogate is trained to map eight measurable plant state variables (flux detector readings, coolant temperatures at key nodes, steam generator pressures, TES and SOE unit states) to core safety indicators: power distribution, PCT, MDNBR, and natural circulation flow rate [17]. Second, the trained surrogate is embedded in the plant process computer, interfaced with the Distributed Control System (DCS) via a read-only data bus with 10-second polling. Alerts are generated when any safety parameter approaches within 10% of its design limit; automated pre-emptive recommendations are generated when a limit is predicted to be breached within 30 minutes. The I&C safety layer is air-gapped from external networks; a one-way data diode separates safety and non-safety layers [10, 11].

The Gaussian Process (GP) surrogate inherently provides a posterior variance



estimate, which is propagated to all predicted safety indicators. This uncertainty is displayed in the control system as a confidence band alongside nominal predictions. When predictive uncertainty increases (e.g., during off-nominal transients outside the training dataset), alert thresholds are adaptively widened to avoid false positives while maintaining conservative safety margins.

4.2. Feasibility

The Aegis-40 methodology is feasible under the team's current conditions for four reasons. First, both OpenMC and OpenFOAM are freely available under open-source licences, eliminating software cost barriers entirely. Second, all codes run on commodity Linux workstations and university HPC clusters, both of which are available to the team at New Uzbekistan University. Full-core OpenMC criticality calculations at the resolution described above are expected to complete in 2–4 hours per burnup step on a 32-core workstation; the depletion sweep over 36 steps therefore requires approximately 3–5 days of wall-clock time, well within the available schedule. OpenFOAM steady-state simulations on the medium mesh (~5 million cells) are expected to converge in 6–12 hours on the same hardware. Third, the CAREM-25 reference design is fully documented in open IAEA publications [1], providing validated baseline parameters from which the Aegis-40 departs incrementally, limiting the risk of fundamental feasibility gaps. Fourth, the NuScale SMR benchmark of Ez Aldeen et al. [3] and the SMART reactor validation studies of Oliveira et al. [21] and Zhang and Sanchez-Espinoza [19] provide external reference data against which the Aegis-40 models can be benchmarked before any design-specific conclusions are drawn, ensuring that numerical errors are identified and corrected before they propagate to the final design.

5. ORIGINALITY, INNOVATION AND LOCALIZATION OF THE PROJECT

5.1. Originality/Innovative Aspects:

The Aegis-40 design distinguishes itself from existing integral Pressurized Water Reactors (iPWRs) through several key architectural and methodological innovations that address the specific energy needs and safety requirements of modular deployment.

- **Integrated Hybrid Energy System:** While traditional SMRs focus primarily on electricity, the Aegis-40 integrates a dual-mode non-electrical application suite. The integration of TES and SOE represents a system-level innovation, enabling continuous reactor operation while serving variable heat and hydrogen demand.
- **Soluble-Boron-Free (SBF) Core:** The Aegis-40 core is designed for full SBF operation using **gadolinia (Gd_2O_3)** burnable absorbers. This eliminates the complex and waste-heavy Chemical and Volume Control System (CVCS)

required by conventional PWRs, reducing radioactive liquid waste and simplifying the plant's auxiliary infrastructure.

- **Physics-Informed Digital Twin:** A novel Instrumentation and Control (I&C) dimension is introduced through a **Digital Twin surrogate model**. By training a Gaussian process regressor on high-fidelity OpenMC and OpenFOAM data, the system can perform real-time safety margin inference and predict potential limit breaches 30 minutes in advance, a capability not present in current reference designs.
- **Quantified Innovation (wFOM):** The project employs a **Weighted Figure of Merit (wFOM)** framework derived from the Analytical Hierarchy Process (AHP) to mathematically justify design choices. This allows for an objective comparison against the CAREM-25 reference, demonstrating targeted improvements such as a 25% increase in electric output and a 48-hour extension in passive safety duration (targeting ≥ 100 hours).

The FOM is defined as:

$$wFOM = \sum_{i=0}^n w_i \cdot \ln\left(\frac{x_{i,design}}{x_{i,ref}}\right)$$

Where $\mathbf{x}_i$ represents the normalized performance ratios and $\mathbf{w}_i$ represents the weight coefficients determined via the **Analytical Hierarchy Process (AHP)**. The team has optimized the weights as follows to prioritize the core objectives of the competition:

Parameter	Variable ($\mathbf{x}_i$)	Weight ($\mathbf{w}_i$)	Rationale
Grace period (autonomous safety)	Δt_{grace}	0.40	Highlights the 120-hour (5-day) autonomous safety window.
Passivity (no active components)	$1+N_{active}$	0.30	Penalizes reliance on active components/external power.
Safety margin (PCT limit)	M_{PCT}	0.15	Ensures Peak Cladding Temperature stays within safety limits.
System volume (footprint)	V_{sys}	0.10	Rewards modularity and spatial efficiency.
Diversity (physical principles)	D_p	0.05	Values the use of multiple physical principles (e.g., Gravity + Evaporation).

Table 1. AHP-weighted FOM parameters to Aegis-40 targets.

By applying this model, the Aegis-40 achieves a positive FOM relative to the reference reactor (CAREM-25), demonstrating a mathematically verified



improvement in safety-to-volume efficiency. This approach allows for a rigorous sensitivity analysis, ensuring that the design remains robust across various accident scenarios.

The wFOM framework is a preliminary evaluation tool, and the assigned weights represent initial engineering judgment. Final weighting will be determined through formal AHP pairwise comparison and validated against full-system simulation results. Therefore, reported FOM values should be interpreted as directional indicators rather than definitive performance metrics.

5.2. Localization Aspects

Turkey's mineral resource base has direct relevance to the Aegis-40 fuel design. The Beylikova formation hosts the world's second-largest rare earth element reserve (694 million tons of ore, 10 of 17 known REEs), including gadolinium, which is the burnable absorber material (Gd_2O_3) central to the Aegis-40's soluble-boron-free core design [6]. ETİ MADEN operates a pilot plant at the site processing 1,200 tons of ore annually, with plans for industrial-scale production of 570,000 tons per year [31, 32]. The location of gadolinium in a domestically processed deposit creates a prospective indigenous supply chain for the current reactivity control strategy.

Additionally, the Aegis-40 is well designed for deployment in Turkey, a country that imports 75% of its primary energy and targets 5 GW of SMR capacity by 2050 [27, 29]. Unlike the Akkuyu build-own-operate model, the Aegis-40 is conceived, simulated, and validated entirely on open-source codes (OpenMC, OpenFOAM), ensuring the host country retains full intellectual property over the design without proprietary licence dependencies [13, 18].

The reactor's 40 MWe scale matches Turkey's grid and geography. Coastal deployment at Sinop — ranked as the optimal Turkish NPP site by multi-criteria analysis [26] — offers a 5°C cooling water advantage over Mediterranean sites [27]. Inland deployment in central Anatolian cities (Eskişehir, Ankara, Sivas) leverages the sorption-based TES system to displace imported natural gas that currently supplies 41% of building heating demand [30].

Turkey's nuclear workforce (trained through Akkuyu), institutional anchors (TENMAK, TÜNAŞ, NDK), and heavy steel manufacturing capacity collectively provide the industrial base for modular SMR fabrication and deployment [27, 29].

6. PROJECT WORK AND TIME MANAGEMENT

The project is divided into several work packages (WPs), progressing from design baseline establishment through detailed analysis to final reporting. The schedule is designed to meet the April 24, 2026 PER deadline and the subsequent Final Evaluation Report (FER) submission.

Time	Work Package	Description
March	Conceptual Foundations	Activities focus on conducting a comparative literature review of existing SMRs to inform our design space and assigning roles for core design, thermal-hydraulics, and 3S compliance.
April	Preliminary Engineering	The team will define the technical baseline for the detailed design, using existing data as a reference, and adding the innovative approaches and solutions.
May–June	Advanced Multiphysics Simulations	The team will utilize OpenFOAM for multiphysics thermal-hydraulic analysis and OpenMC for depletion and burnup tracking. The CAD model will be designed and implemented.
June–July	Infrastructure Integration	Developing a thermodynamic model for the sorption-based energy storage system and Hydrogen system. Designing the NPP layout and I&C.
June–August	Development (ML + FOM)	Development of the Weighted Logarithmic FOM using Machine Learning to justify design choices and optimize parameters for safety, economic feasibility, and sustainability.
August	Finalization & Benchmarking	The team will finalize and benchmark all technical analyses and prepare the final presentation files.
September	Competition Presentation	The team will participate in the competition final presentations at the TENMAK Istanbul campus.

Table 2. Project work schedule divided into work packages.

7. BROADER IMPACTS AND TARGET AUDIENCE

7.1. Broader Impacts:

Each 40 MWe Aegis-40 unit delivers baseload low-carbon electricity to approximately 40 000–60 000 Turkish households, displacing natural-gas generation ($\approx 430 \text{ g CO}_2/\text{kWh}$) and avoiding about 5.6 million tonnes CO_2 over 40 years. This supports Turkey’s 2053 net-zero goal and reduces exposure to gas-price volatility.

The sorption-based TES co-generation system provides $\approx 15 \text{ MWth}$ of district heat for 7 000–10 000 households, offsetting part of the 41 % of national gas use in building heating and saving several hundred million ₺ per unit annually. Cold-climate coastal cities — Sinop, Kastamonu, Samsun — are key beneficiaries.

Night-time SOE operation yields $\approx 120 \text{ t}$ of green hydrogen per year per unit at costs competitive with off-peak grid electrolysis, serving Sinop’s industry, refuelling network, and on-site backup systems. Deploying 5–10 units would advance Turkey’s hydrogen economy and European Hydrogen Partnership goals.



Scientifically, the project delivers open-source OpenMC and OpenFOAM models for 40 MWe iPWRs, a Digital Twin surrogate, and SOE-PWR integration methodology — filling major gaps in SMR research for Turkey and Central Asia.

7.2. Target Audience:

The primary target audience is Turkey's nuclear energy establishment: the Nuclear Regulatory Authority (NDK), the Turkish Energy, Nuclear and Mining Research Institute (TENMAK), the state generation company EUAS and its Sinop-focused subsidiary TUNAS, and the Ministry of Energy and Natural Resources. These are the institutions that will evaluate whether a 40 MWe iPWR class has a complementary role alongside the large-unit Akkuyu and Sinop programmes. The AHP-weighted FOM evaluation framework and the openly documented methodology are specifically designed to support the type of comparative technology assessment these bodies perform.

Secondary audiences include: Turkish district heating network operators and municipal energy companies in Black Sea and eastern Anatolian cities seeking low-carbon heat sources to replace natural gas; industrial energy users and hydrogen off-takers interested in the SOE co-production pathway; the international SMR design and licensing community working on IAEA regulatory harmonisation frameworks; and universities and research institutes in Turkey and Uzbekistan developing nuclear engineering education programmes.

8. DESIGN PREPARATION PHASE

This section presents the pre-design preparation for the Aegis-40, including reference reactor characterisation, key design parameter comparisons with three iPWR references, and the computational tools to be employed throughout the project.

8.1. Reference Reactors

Three iPWR designs serve as references for the Aegis-40. The CAREM-25 (32 MWe, Argentina) is the closest power-class match and primary FOM reference, employing natural circulation and a fully documented open IAEA parameter set [1]. The SMART reactor (107 MWe, South Korea) provides benchmark neutronics and thermal-hydraulic data through the validated OpenFOAM study of Oliveira et al. [21] and the OpenMC SMR benchmark of Ez Aldeen et al. [3], directly applicable to Aegis-40 model validation. The NuScale VOYGR (70 MWe/module) provides the regulatory precedent for site-boundary EPZ, below-grade pool placement, and passive cooling duration targets [2, 3].

8.2. Basic Design Parameters Table

Table 3 presents key design parameters of the three reference reactors alongside the Aegis-40 target parameters.

Parameter	CAREM-25	SMART	NuScale VOYGR	Aegis-40	Comparison
Reactor type	iPWR	iPWR	iPWR	iPWR	All are iPWRs with compact, modular containment.
Thermal power	~100 MWth	540 MWth	~250 MWth	~125 MWth	Aegis-40 in mid-range; balanced for hybrid uses.
Electric power	~32 MWe	107 MWe	~77 MWe	40 MWe	Medium output, ideal for distributed generation.
Coolant	H ₂ O	H ₂ O	H ₂ O	H ₂ O	Proven reliability
Primary circulation	Natural circ.	Forced pumps	Natural circ.	Natural circ.	Passive natural circ. like CAREM and NuScale.
Fuel material	UO ₂	UO ₂	UO ₂	UO ₂	Identical UO ₂ fuel simplifies modeling and supply.
Enrichment	~3.1%	~4.8%	~4.95%	~4.0%	Moderate; optimized for burnup and safety margin.
Fuel cycle	14 months	36 months	24 months	18 months (target)	Intermediate
Reactivity control	Rods + boron	Rods + boron	Rods only	Rods + Gd ₂ O ₃ (SBF)	Uses Gd ₂ O ₃ absorbers, avoiding soluble boron.
Passive safety duration	~72 h (est.)	~72 h	≥72 h (pool)	≥100 h (target)	Extended > 100 h autonomous cooling capability.
Target burnup	~33 GWd/MTU	~60 GWd/MTU	~62 GWd/MTU	~45 GWd/MTU avg.	Intermediate

Parameter	CAREM-25	SMART	NuScale VOYGR	Aegis-40	Comparison
EPZ boundary	Site boundary	Site boundary	Plant boundary	Plant boundary (target)	Increased Safety and Security
Non-electric al app.	Desalination , process heat	Process heat, desalination	District heating (TES)	TES heat + H ₂ (SOE)	Most versatile
Neutronics code	INVAP (prop.)	HELIOS/MAS TER	CASMO / SIMULATE	OpenMC (open)	Transparent, collaborative validation.
TH code	RELAP5	MARS	NRELAP5	OpenFOAM (open)	Flexible transient CFD coupling.
I&C / monitoring	Analogue/digital	Digital DCS	Digital DCS	Digital DCS + Digital Twin surrogate	Advanced DCS with predictive digital-twin analytics.

Table 3. Basic Design Parameters: Three Reference Reactors vs. Aegis-40.

8.3. Comparative Assessment

The Aegis-40 targets a power level between CAREM-25 and NuScale VOYGR, filling a gap in the 40 MWe range that is underserved by current designs. Relative to CAREM-25, the Aegis-40 offers: higher electric output (+25%), longer fuel cycle (+4 months), soluble-boron-free operation, extended passive safety duration (+48 hours target), and dual non-electrical applications (TES + SOE hydrogen) absent from CAREM-25. Relative to SMART and NuScale VOYGR, the Aegis-40 offers a significantly smaller footprint and capital cost appropriate to Turkey's Black Sea coastal grid and city-scale district heating loads. The Digital Twin surrogate I&C architecture is not present in any of the three reference designs and represents the most novel operational dimension of the Aegis-40 concept.

8.4. Computer Codes and Software

The following open-source and standard tools are planned for the Aegis-40 design process: OpenMC v0.15+ (Monte Carlo neutronics and depletion, Python/C++); OpenFOAM v9+ (CFD thermal-hydraulics, C++); foamForNuclear platform (two-phase TH solvers); FreeCAD and Salome (CAD geometry and mesh generation); Python scientific stack with scikit-learn (Gaussian process surrogate model training for Digital Twin); OpenMC sensitivity toolkit (nuclear data uncertainty quantification); and standard Python libraries (NumPy, SciPy, Matplotlib) for post-processing. No proprietary licences are required for any component of the computational toolchain, consistent with the indigenization objective.



Design Phase	Code / Tool	Language / Platform	Planned Use
Phase 1 – Neutronics	OpenMC v0.15+	Python / C++	k-eff criticality, 3D flux tallies, depletion (CRAM45), reactivity coefficients, sensitivity analysis
Phase 1	ENDF/B-VIII.0 library	HDF5 data files	Continuous-energy cross sections for all isotopes and temperatures
Phase 2 – TH	OpenFOAM v9+	C++	Steady-state nat. conv. (buoyantSimpleFoam), conjugate HT, transient station blackout and SLB (buoyantPimpleFoam)
Phase 2	foamForNuclear	C++ (OpenFOAM lib.)	Euler-Euler two-phase flow solvers for boiling margin analysis
Phase 2	AutoCAD/Fusion360/FreeCAD	Python / GUI	CAD geometry construction and mesh generation
Phase 3, 4 – TES/SOE	Python (NumPy/SciPy) and other TBD	Python 3	Zeolite TES energy balance, SOE hydrogen yield model, daily demand-matching optimisation
Phase 5 – Digital Twin	scikit-learn (GaussianProcessRegressor)	Python 3	GP surrogate model training on OpenMC/OpenFOAM simulation dataset; real-time safety margin inference
All phases	NumPy / SciPy / Matplotlib	Python 3	Post-processing, visualisation, statistical analysis, uncertainty propagation
All phases	Git + GitHub	Cross-platform	Version control, team collaboration, reproducibility of all simulation results

Table 4. Computational tools planned for each design phase of the Aegis-40 project.



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